

ZFS utilities as shown below:

```
apt install zfsutils-linux
```

In my set-up, I have `/dev/sda` as the OS drive and `/dev/sdb` as another drive, on which I'll be creating the ZFS pool. ZFS is very sensitive to the disk names provided when creating the pool; so to ensure that it picks up the proper disk, we need to go to `/dev/disk/by-id` to find the appropriate disk.

```
root@ubuntu:~# ls -lh /dev/disk/by-id
total 0
lrwxrwxrwx 1 root root 9 Dec 4 10:55 scsi-14d5346542029202
07305e3437703544694957d7ced624a7d -> ../../sr0
lrwxrwxrwx 1 root root 9 Dec 4 10:55 scsi-3600224804e58652
893517c552e9d9cc3 -> ../../sdb
lrwxrwxrwx 1 root root 9 Dec 4 10:55 scsi-360022480cf72d7b
58630bab61658ad98 -> ../../sda
lrwxrwxrwx 1 root root 10 Dec 4 10:55 scsi-360022480cf72d7b
58630bab61658ad98-part1 -> ../../sda1
lrwxrwxrwx 1 root root 10 Dec 4 10:55 scsi-360022480cf72d7b
58630bab61658ad98-part2 -> ../../sda2
lrwxrwxrwx 1 root root 10 Dec 4 10:55 scsi-360022480cf72d7b
58630bab61658ad98-part3 -> ../../sda3
lrwxrwxrwx 1 root root 9 Dec 4 10:55 wwn-
0x600224804e58652893517c552e9d9cc3 -> ../../sdb
lrwxrwxrwx 1 root root 9 Dec 4 10:55 wwn-0x60022480cf72d7b
58630bab61658ad98 -> ../../sda
lrwxrwxrwx 1 root root 10 Dec 4 10:55 wwn-0x60022480cf72d7b
58630bab61658ad98-part1 -> ../../sda1
lrwxrwxrwx 1 root root 10 Dec 4 10:55 wwn-0x60022480cf72d7b
58630bab61658ad98-part2 -> ../../sda2
lrwxrwxrwx 1 root root 10 Dec 4 10:55 wwn-0x60022480cf72d7b
58630bab61658ad98-part3 -> ../../sda3
```

As seen above in the output, the disk-ID for `sdb` is `scsi-3600224804e58652893517c552e9d9cc3`. If you use partitions on an existing disk, take the ID numbers from `/dev/disk/by-id`. If you use normal names like `/dev/sda` or `/dev/sdb`, then you may not be able to get back the ZFS pool on reboot, in case the order of disks changes for some reason – which will cause the names of the disks to be changed.

```
root@ubuntu:~# zpool create zdata /dev/disk/by-id/scsi-
3600224804e58652893517c552e9d9cc3
```

```
invalid vdev specification
use '-f' to override the following errors:
/dev/disk/by-id/scsi-3600224804e58652893517c552e9d9cc3 does
not contain an EFI label but it may contain partition
```

And there, I got an error while creating the pool. Since I know that the disk doesn't contain anything, I'll simply

force it using `-f` and create my pool.

```
root@ubuntu:~# zpool status
pool: zdata
state: ONLINE
scan: none requested
config:
```

NAME	STATE	READ	WRITE	CKSUM
zdata	ONLINE	0	0	0
scsi-3600224804e58652893517c552e9d9cc3	ONLINE	0	0	0

```
errors: No known data errors
```

This is a very basic ZFS pool with just one disk. ZFS directly supports mirroring, striping and RAID5 modes. So it eliminates the need for MDADM or LVM. Documentation of such advanced configurations is easily available online; in particular, FreeBSD's ZFS guide is very easy to understand and apply the information to Ubuntu in spite of it being so different from FreeBSD.

Now we can initialise LXD using the following command:

```
lxd init
```

It will ask you a few questions and offer to even create a ZFS pool, but I have not used that option since I always create my ZFS pools manually, which gives me flexibility on how to set them up.

```
root@ubuntu:~# lxd init
```

```
Name of the storage backend to use (dir or zfs) [default=zfs]:
Create a new ZFS pool (yes/no) [default=yes]? no
Name of the existing ZFS pool or dataset: zdata
Would you like LXD to be available over the network (yes/no)
[default=no]?
Do you want to configure the LXD bridge (yes/no) [default=yes]?
Warning: Stopping lxd.service, but it can still be activated
by:
    lxd.socket
LXD has been successfully configured.
```

By default, `lxdbr0`, the interface that is created by this setup script will do a NAT. In case you don't want to do a NAT, you can either select that option during the init time or run `dpkg-reconfigure lxd` after this to do it again.

Creating your first container

This is pretty simple. First, use the following command:

```
lxd launch ubuntu: first-container
```

In the above command, `launch` is the command name for creating and launching a container; `ubuntu` is the name